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Predicting In-Hospital Maternal Mortality and Severe Maternal Morbidity in MIMIC-IV with a Stacked Machine Learning Ensemble under Extreme Class Imbalance

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Highlights

What are the main findings?

- Maternal death is too rare in MIMIC-IV to model reliably (41 of 25,736 obstetric admissions, 0.16%): five-fold cross-validated AUC for the stacked ensemble was 0.871 $\pm$ 0.050 but average precision was only 0.162 $\pm$ 0.089.
- Using severe maternal morbidity (SMM; 475 positives, 1.85%) as the outcome gave a markedly more stable task: on a held-out test set the stacked ensemble reached an AUC of 0.913 and an average precision of 0.384 (about 21 times the prevalence), with a recall of 0.79.

What are the implications of the main findings?

- In single-center data, SMM is a more suitable modelling target than mortality for maternal risk prediction; point estimates for mortality should not be over-interpreted.
- Oversampling (SMOTE) did not improve precision-based metrics, and outlier winsorization avoided discarding 19.5% of deaths and 38.7% of SMM cases that row deletion would have removed.

Abstract

Background: Maternal mortality remains a global health priority, but deaths are so rare in single-center data that standard machine learning (ML) evaluation is unreliable. Severe maternal morbidity (SMM), defined by CDC/HRSA diagnosis and procedure indicators, is about one hundred times more frequent and is widely used as a near-miss proxy. **Methods**: We built two models on the MIMIC-IV electronic health record database from 25,736 obstetric hospital admissions of women aged 12–55 years: Model A (death recorded during or after the index admission; 41 events, 0.159%) and Model B (at least one of 20 CDC/HRSA FAD-2023 SMM indicators; 475 events, 1.846%). The pipeline included sentinel-value and implausible-zero cleaning, interquartile-range (IQR$\times$3) winsorization instead of row deletion, skewness-driven imputation, differential scaling, and expert-grouped diagnosis indicators (used in Model A only, to avoid outcome leakage in Model B). Random Forest, XGBoost, Extra Trees, Logistic Regression and a stacked ensemble were tuned by cross-validated grid search on average precision. Stratified five-fold cross-validation was the primary evidence for Model A, and SMOTE was assessed as a sensitivity analysis. **Results**: For Model A, the stacked ensemble achieved a cross-validated AUC of 0.871 $\pm$ 0.050 and average precision (AP) of 0.162 $\pm$ 0.089 (prevalence 0.0016); the single hold-out test contained only eight deaths. For Model B, the stacked ensemble obtained the best hold-out

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AUC (0.913) and AP (0.384; prevalence 0.018) with recall 0.79 and precision 0.14 at the default threshold; restricted to delivery hospitalizations, AP was 0.646. SMOTE did not improve AP. **Conclusions**: In this single-center cohort, SMM supports considerably more stable ML evaluation than mortality. Results remain exploratory, because the SMM label is ICD-based, the data are from one institution, and no external validation was performed.

Keywords: maternal mortality; severe maternal morbidity; near miss; MIMIC-IV; stacked ensemble; class imbalance; precision–recall; machine learning; electronic health records

1. Introduction

Maternal mortality remains a major public health problem: the World Health Organization and partner agencies estimated roughly 287,000 maternal deaths worldwide in 2020 [1]. In high-resource hospitals, however, maternal death is a very rare event. This rarity is a statistical obstacle for supervised learning: with a prevalence of a few cases per thousand admissions, individual cross-validation folds and test sets contain only a handful of positives, so that a single misclassified patient can change recall by several percentage points, and accuracy and ROC-AUC alone become uninformative [2,3].

A widely used way to mitigate this problem is to model *severe maternal morbidity* (SMM), or "near miss", as a surrogate outcome. The U.S. Centers for Disease Control and Prevention (CDC) define SMM through a set of ICD-10-CM diagnosis and ICD-10-PCS procedure indicators, which the Health Resources and Services Administration (HRSA) distributes in its Federally Available Data (FAD) resource document [4,5]. SMM events occur approximately one hundred times more frequently than in-hospital maternal deaths among delivery and postpartum hospitalizations [6], and administrative diagnosis codes have been shown to carry prognostic information for maternal morbidity [7]. The trade-off is label quality: a validation against medical-record review at a public hospital reported low sensitivity and positive predictive value for ICD-10-based SMM identification [8].

The MIMIC-IV database, from Beth Israel Deaconess Medical Center, offers de-identified hospital-wide records with laboratory results, charted vital signs, and diagnosis codes [9,10]. Although MIMIC-IV is not an obstetric database, it contains thousands of admissions of women with pregnancy-related diagnoses, which allows a reproducible comparison of the two outcome definitions on the same patients and the same predictors.

Ensemble stacking combines heterogeneous base learners through a meta-learner and often outperforms its components [11,12]. Whether this advantage persists under extreme imbalance, and whether common remedies such as oversampling help [13], is an empirical question. This study therefore makes the following contributions:

- A side-by-side comparison of two outcomes on one obstetric cohort of 25,736 MIMIC-IV admissions: in-hospital-recorded maternal death (Model A) and CDC/HRSA-defined SMM (Model B).
- An imbalance-aware evaluation protocol: probability-based ROC and precision–recall analysis, average precision as the tuning criterion, stratified cross-validation as the primary evidence for the rarest outcome, and an explicit SMOTE sensitivity analysis.
- Explicit leakage control for SMM: diagnosis-group indicators derived from the same ICD codes that define the outcome are excluded from Model B, and a sensitivity analysis restricts evaluation to delivery hospitalizations (the CDC-comparable denominator).
- A winsorization-based outlier policy that retains every positive case.

The remainder of the article is organized as follows. Section 2 describes data and methods, Section 3 presents the results, Section 4 discusses findings and limitations, and Section 5 concludes.

2. Materials and Methods

2.1. Data source and study cohort

We used MIMIC-IV hosted on PostgreSQL [9] ([**TODO:** state the exact MIMIC-IV version, e.g., v2.2 or v3.1, and the PhysioNet access date]). The study cohort comprises hospital admissions of women aged 12–55 years with at least one obstetric or pregnancy-related ICD diagnosis (diagnosis position $\leq$ 4), identified by an expanded diagnosis universe combining text matching on the ICD long titles with ICD ranges. An intensive care unit (ICU) stay was *not* required: this is important because 30 of the 41 recorded deaths (73%) occurred in admissions without any ICU stay. The final cohort contains 25,736 admissions, of which 8,857 were identified as delivery hospitalizations. Figure 1 summarizes the workflow.

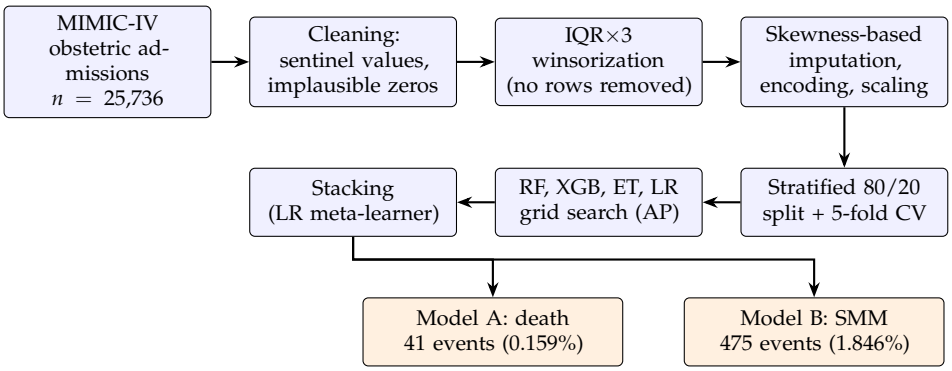


Figure 1. Overview of the pipeline. The same cleaning, preprocessing and learning steps are applied to both outcomes.

2.2. Outcome definitions

Model A (mortality). The outcome is 1 if a hospital death time was recorded during the index admission or in any later admission of the same patient, and 0 otherwise. This yielded 41 events among 25,736 admissions (0.159%).

Model B (SMM). An admission is SMM-positive if it carries at least one ICD-10-CM diagnosis or ICD-10-PCS procedure code matching any of the 20 currently active CDC/HRSA FAD-2023 indicators [4,5,14] (the former indicator on blood-product transfusion, retired by HRSA in 2023, is excluded). The indicators are grouped into seven clinical categories (cardiac, renal, respiratory, hemorrhage, sepsis, other obstetric, other medical). There were 475 SMM-positive admissions (1.846%); among the 8,857 delivery hospitalizations, 252 were positive (2.845%, i.e., 284.5 per 10,000), above the published reference range of 58–88 per 10,000 deliveries [14]. The higher rate is expected, as MIMIC-IV is a tertiary referral center and the expanded denominator also includes non-delivery admissions. The most frequent indicators were acute renal failure (180 admissions), shock (72), ARDS (66), sepsis (62), and disseminated intravascular coagulation (61); 23.4% of positive admissions met indicators from more than one clinical category.

2.3. Predictors and preprocessing

Candidate predictors comprised administrative variables (age at admission, ethnicity group, insurance, language group, admission type, admission location, ICU stay and ICU stay count) and the earliest available value of 27 chart/laboratory variables (vital signs, Glasgow Coma Scale components, hematology and chemistry). For the 13 laboratory

variables, values were also taken from hospital-wide laboratory events (not only from ICU chart events), which matters here because most deaths occurred outside the ICU. Preprocessing steps were:

1. **Sentinel values and implausible zeros.** The placeholder 999,999 and physiologically impossible zeros were set to missing.
2. **Outlier handling by winsorization.** For each continuous variable, values beyond $Q_1 - 3\,\mathrm{IQR}$ or $Q_3 + 3\,\mathrm{IQR}$ were clipped to the nearest bound (with clinical lower bounds), so that no admission is deleted. Ordinal scales and variables with $\mathrm{IQR} = 0$ were not clipped. Our first implementation deleted rows with out-of-bound values, as in our previous neonatal pipeline; an audit showed that this removed 8 of 41 deaths (19.5%) in Model A and 184 of 475 SMM cases (38.7%) in Model B, so it was replaced by winsorization.
3. **Imputation.** Median for variables with $|\text{skewness}| > 1$, mode otherwise.
4. **Encoding and scaling.** One-hot encoding of the five categorical groups; `RobustScaler` for variables with $|\text{skewness}| > 1$ and `MinMaxScaler` otherwise. Near-constant ICU-monitoring variables (motor GCS and Richmond Agitation–Sedation Scale; the most common value covered more than 99% of rows) were dropped from Model A.

Diagnosis-group indicators and leakage control. ICD diagnosis codes in the cohort (4,116 distinct codes for Model A) were aggregated into 26 binary diagnostic group indicators (`gdg_m01`–`gdg_m26`) by clinical classification ([**TODO:** describe the classification procedure and the panel composition, or state that it was performed by the authors]). Two further indicators reserved for residual clustering were constant and removed. Because these indicators are derived from the same coded diagnoses that define SMM, they were *excluded* from Model B; likewise the seven per-category SMM counts, the multiclass label, and the delivery-hospitalization flag were not used as predictors (the latter is kept only to define the evaluation subset). Model A retained the 26 indicators, giving 89 predictors; Model B used 65 predictors. This difference in the predictor sets is considered in Section 4.

2.4. Learning algorithms and evaluation

We trained Random Forest (RF), XGBoost (XGB), Extra Trees (ET) and L2-regularized Logistic Regression (LR) on an 80/20 stratified split (random seed 42): 20,588 training and 5,148 test admissions, with 33 and 8 deaths in Model A and 380 and 95 SMM cases in Model B. Imbalance was handled through class weighting (`balanced` or `balanced_subsample`; XGB `scale_pos_weight` of 622.9 in A and 53.2 in B). Hyperparameters were tuned with grid search using stratified five-fold cross-validation on the training set with *average precision* as the criterion; grids were deliberately small for Model A. The tuned learners were combined into a stacked ensemble [11] with a class-weighted L2 logistic-regression meta-learner whose inputs were generated by the same five-fold stratified scheme. Implementation used scikit-learn, XGBoost and imbalanced-learn [15–17] with RF [18] and ET [19] base models.

ROC and precision–recall (PR) curves and the AUC and AP were computed from predicted probabilities of the positive class, while the classification report and confusion matrix used the default 0.5 decision threshold. Since only 8 deaths fall in the Model A test set, the main evidence for Model A is stratified five-fold cross-validation over all 25,736 admissions, reported as mean $\pm$ standard deviation for AUC, AP, recall, F1 and balanced accuracy [20]. The cross-validation re-fits the already tuned configurations on the full data set; its estimates are therefore slightly optimistic, as the hyperparameters were selected on the training split. As a sensitivity analysis, SMOTE (sampling strategy 0.2, $k = 5$) was placed inside each training fold of a pipeline to avoid leakage into the validation folds. For Model B, the stacked ensemble was additionally evaluated on the delivery-hospitalization

subset of the test set, without re-training. The study follows the TRIPOD+AI reporting guidance where applicable [21] ([**TODO:** attach the TRIPOD+AI checklist as supplementary material]).

3. Results

3.1. Cohorts and outcomes

Table 1 compares the two prediction tasks. Model B has 11.6 times more events than Model A and 95 rather than 8 positives in the test set.

Table 1. Comparison of the two prediction tasks (same 25,736 admissions).

	Model A (mortality)	**Model B (SMM)**
Positive admissions	41 (0.159%)	475 (1.846%)
Positives in training / test split	33 / 8	380 / 95
Predictors	89 (incl. 26 diagnosis groups)	65 (no diagnosis groups)
Weight for the positive class (XGBoost)	622.9	53.2
Positives retained after outlier handling	41 of 41	475 of 475

3.2. Model A: mortality

Table 2 and Figure 2 show the cross-validated results. The stacked ensemble had the highest AUC (0.871 $\pm$ 0.050) and the highest balanced accuracy and recall (0.771 and 0.661), but its F1 score was only 0.017, and its AP (0.162 $\pm$ 0.089) was comparable to that of XGB and ET and far below any clinically useful level in absolute terms, although it is roughly one hundred times the prevalence of 0.0016. Fold-to-fold variability was large (standard deviations of AP between 0.057 and 0.105) and RF produced no positive predictions at the default threshold (recall 0). On the hold-out test set (8 deaths), AUC values ranged from 0.745 (RF) to 0.878 (LR), with the stacked ensemble at 0.866 and AP of 0.129 (Table 4). Its confusion matrix contained 7 true positives, 1 false negative, 624 false positives and 4,516 true negatives (recall 0.88, precision 0.011); these point estimates rest on eight patients and must be read as illustrative only.

Table 2. Model A: stratified five-fold cross-validation (mean $\pm$ SD) on all 25,736 admissions (41 deaths).

Model	**AUC**	**AP**	**Recall**	**F1**	**Bal. acc.**
Random Forest	0.857 $\pm$ 0.065	0.118 $\pm$ 0.057	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.500 $\pm$ 0.000
XGBoost	0.775 $\pm$ 0.070	0.159 $\pm$ 0.094	0.167 $\pm$ 0.115	0.172 $\pm$ 0.107	0.583 $\pm$ 0.057
Extra Trees	0.846 $\pm$ 0.077	**0.173 $\pm$ 0.105**	0.075 $\pm$ 0.100	0.124 $\pm$ 0.162	0.537 $\pm$ 0.050
Logistic Regression	0.769 $\pm$ 0.064	0.058 $\pm$ 0.057	0.464 $\pm$ 0.049	0.019 $\pm$ 0.004	0.694 $\pm$ 0.024
Stacking	**0.871 $\pm$ 0.050**	0.162 $\pm$ 0.089	**0.661 $\pm$ 0.136**	0.017 $\pm$ 0.001	**0.771 $\pm$ 0.059**

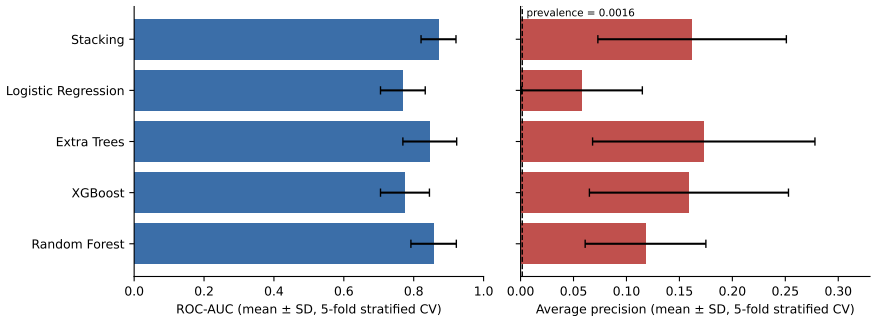


Figure 2. Model A: cross-validated AUC and average precision with standard deviations. The dashed line marks the outcome prevalence (41/25,736).

3.3. Oversampling sensitivity analysis (Model A)

Adding SMOTE inside each training fold increased the AUC of all five models (e.g., stacking 0.871 to 0.883, XGB 0.775 to 0.838) and the recall of most tree-based models, but not that of the stacked ensemble (0.661 to 0.122), and it did not improve AP for the best models: stacking 0.162 to 0.161, ET 0.173 to 0.159, XGB 0.159 to 0.121 (Table 3, Figure 3). The changes in AUC and AP are all smaller than one standard deviation, whereas the recall of the stacked ensemble dropped markedly. Since oversampling mainly shifts the operating point rather than adding information, and AP did not improve, class weighting was kept as the default.

Table 3. Model A: effect of SMOTE (sampling strategy 0.2) on five-fold cross-validated AUC, AP and recall (mean ± SD).

	AUC		AP		Recall	
Model	Weights	SMOTE	Weights	SMOTE	Weights	SMOTE
Random Forest	0.857	0.872	0.118	0.125	0.000	0.097
XGBoost	0.775	0.838	0.159	0.121	0.167	0.197
Extra Trees	0.846	0.858	0.173	0.159	0.075	0.097
Logistic Regression	0.769	0.801	0.058	0.064	0.464	0.169
Stacking	0.871	0.883	0.162	0.161	0.661	0.122

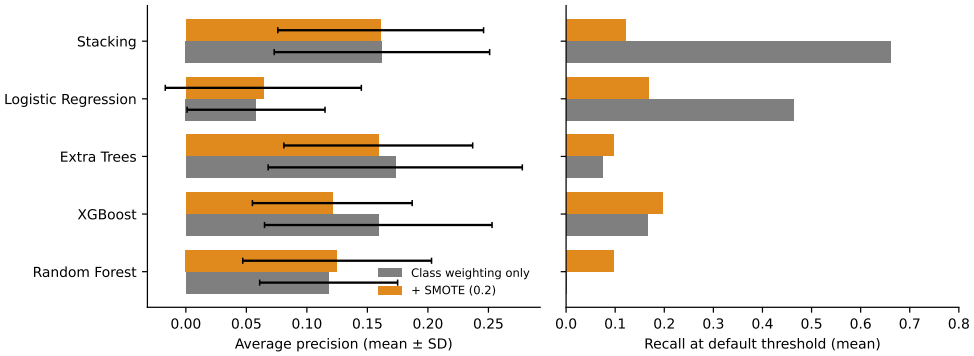


Figure 3. Model A: class weighting versus SMOTE in five-fold cross-validation.

3.4. Model B: severe maternal morbidity

Model B showed a better and much more stable behavior. On the hold-out test set (95 SMM cases), the stacked ensemble was the best model in both AUC (0.913) and AP (0.384), followed by XGB (AUC 0.909, AP 0.358) and RF (AUC 0.900, AP 0.360) (Table 4). The AP of the stacked ensemble is about 21 times the test prevalence (0.018). At the default threshold its confusion matrix showed 75 true positives, 20 false negatives, 473 false positives and 4,580 true negatives (recall 0.79, precision 0.14). Restricted to the 1,802 test admissions that were delivery hospitalizations (58 SMM cases, 3.219%), the same model reached a recall of 0.78, precision of 0.27, F1 of 0.40 for the positive class and an AP of 0.646 (versus 0.384 in the whole test set), with an overall accuracy of 0.93. Figure 4 shows the ROC and PR curves of the stacked ensemble for both models.

Table 4. Hold-out test performance (probabilities of the positive class): AUC and average precision. Model A test set: 5,148 admissions, 8 deaths; Model B test set: 5,148 admissions, 95 SMM cases.

	Model A (mortality)		Model B (SMM)	
Model	AUC	AP	AUC	AP
Random Forest	0.745	0.096	0.900	0.360
XGBoost	0.758	0.171	0.909	0.358
Extra Trees	0.795	0.150	0.900	0.315
Logistic Regression	0.878	0.108	0.889	0.339
Stacking	0.866	0.129	**0.913**	**0.384**

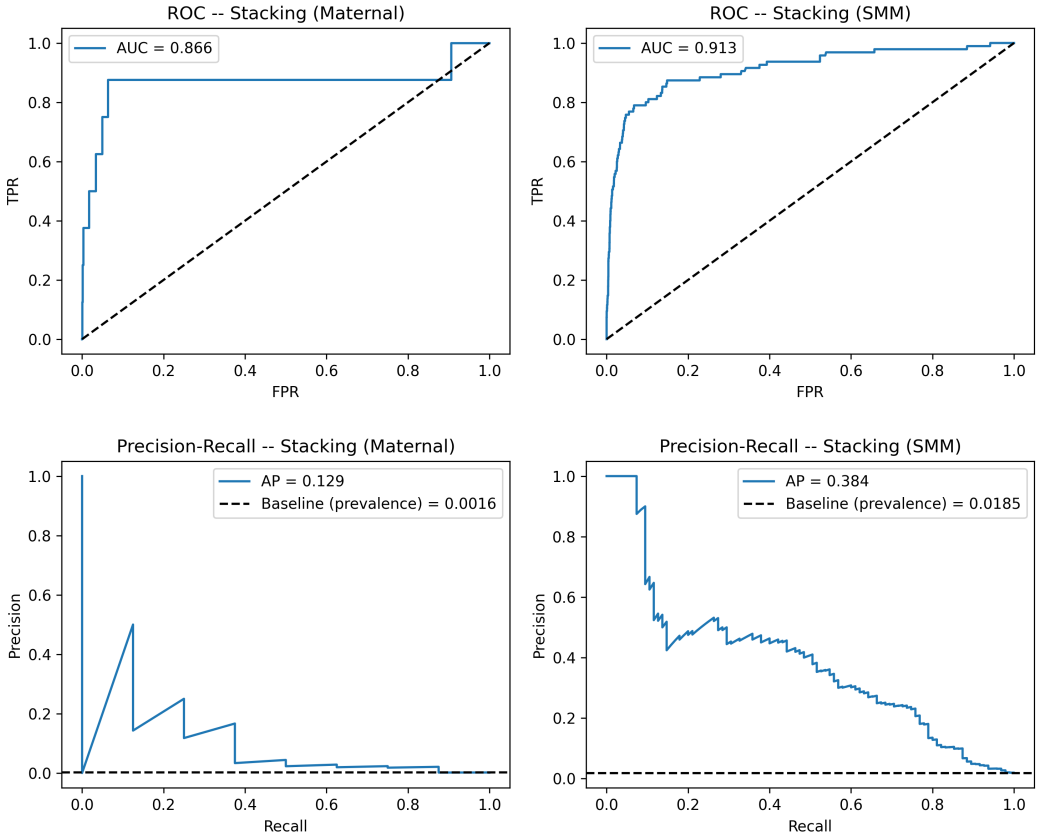


Figure 4. ROC (top) and precision–recall (bottom) curves of the stacked ensemble on the hold-out set. Left: Model A; right: Model B.

3.5. Predictor importance

Figure 5 shows the 30 most important predictors of the tuned Random Forest for each model. For Model A, age at admission ranked first, followed by hematocrit, hemoglobin, the diagnosis-group indicator `gdg_m13`, and creatinine. For Model B, the top positions were occupied by laboratory values: creatinine, glucose, bicarbonate, magnesium, and hemoglobin. Impurity-based importances are descriptive of the fitted forests and not evidence of causal effects; moreover, the importance threshold used for feature screening was computed but not applied, so that all predictors were retained in every model.

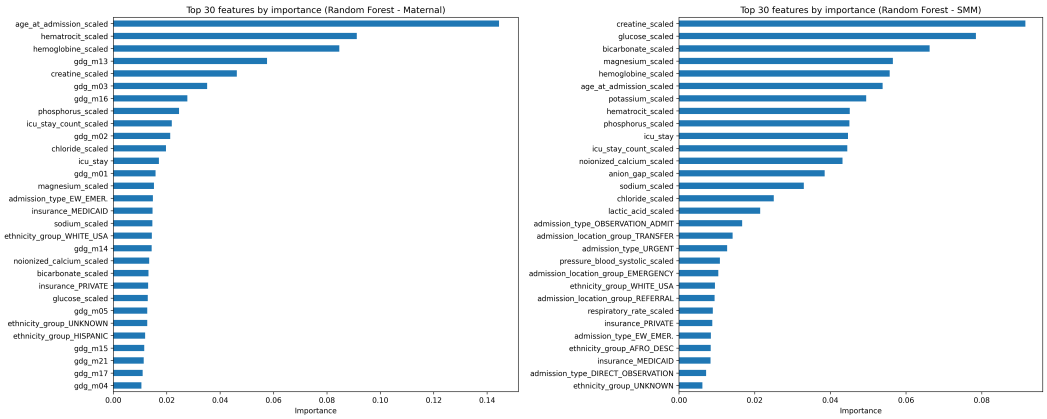


Figure 5. Top 30 Random Forest feature importances. Left: Model A; right: Model B.

4. Discussion

Mortality is not a modelling target at this sample size. With 41 deaths, cross-validated AUC values around 0.85–0.87 coexisted with AP of 0.12–0.17, with standard deviations of the same order as the means, and with hold-out conclusions supported by eight deaths. A ranking of the five algorithms is therefore not reliable: for example, LR achieved the best hold-out AUC (0.878) but the lowest cross-validated AUC and AP (0.769 and 0.058), whereas RF had the lowest hold-out AUC (0.745) but the second-highest cross-validated AUC (0.857). Our results support the recommendation to report precision–recall metrics and fold-level dispersion for rare outcomes [2,3].

SMM is a more stable surrogate, but a noisy one. A positive class eleven times larger produced hold-out AUC values of 0.889–0.913 with a narrow spread across algorithms, a stacked ensemble that was best on both AUC and AP, and a precision–recall trade-off that can be discussed clinically. Two limitations should be stated plainly. First, the SMM label is ICD-based; ICD-10 coding showed low sensitivity and positive predictive value in a medical-record validation study [8], so our positives are an administrative proxy, and the model may partly learn coding and care-intensity patterns. This is also likely to explain the high SMM rate in delivery hospitalizations. Second, the leading predictors of Model B were creatinine, glucose, bicarbonate and magnesium, which relate to renal and metabolic derangement; since acute renal failure is the most frequent indicator (180 admissions), part of the signal reflects laboratory abnormalities that co-occur with the outcome definition rather than anticipate it. Using the earliest available value mitigates but does not remove this concern, and prospective timing of predictors relative to the event was not enforced.

Oversampling and outlier policy. SMOTE did not improve the precision-based metrics, in line with evidence that imbalance corrections can harm calibration and do not necessarily improve discrimination [13]. Replacing row deletion by winsorization preserved all positive cases; retaining them matters more than the geometric tidiness of the feature distribution, since deleting extreme laboratory values removes precisely the sickest patients.

Limitations. (i) Single-center data and no external validation. (ii) Model A includes the 26 diagnosis-group indicators whereas Model B excludes them; the two models are therefore not strictly comparable in predictor sets, and a sensitivity analysis of Model A without these indicators is [**TODO:** pending: re-run Model A without `gdg_m*` and report]. (iii) Imputation values and scalers were fitted before the train/test split, and the cross-validation re-fits hyperparameters tuned on the training split, which can make estimates slightly optimistic. (iv) Model B has no cross-validated estimate over the full data ([**TODO:** add five-fold CV for Model B for symmetry with Model A]). (v) The mortality outcome captures only deaths recorded by the hospital; post-discharge deaths are not observed. (vi) Metrics at the

default 0.5 threshold are shown for transparency; no calibration or threshold optimization was carried out, and predicted probabilities should not be interpreted as absolute risks. (vii) MIMIC-IV is not an obstetric registry and may differ from the population of obstetric services in other health systems. The models are research prototypes and should not guide individual clinical decisions.

5. Conclusions

In 25,736 obstetric admissions of MIMIC-IV, in-hospital-recorded maternal death (41 events) could not be modelled with reliable precision, whereas SMM (475 events) allowed a stable evaluation, with a stacked ensemble reaching a hold-out AUC of 0.913 and AP of 0.384. Class weighting was preferable to SMOTE, and winsorization protected positive cases from deletion. Future work should validate these models on independent cohorts, calibrate and threshold them for a decision-support use case, and evaluate the secondary multiclass task of predicting the SMM category.

Author Contributions: Conceptualization, E.G.J. and J.D.M.D.; methodology, E.G.J.; software, E.G.J.; validation, E.G.J. and J.D.M.D.; formal analysis, E.G.J.; data curation, E.G.J.; writing—original draft preparation, E.G.J.; writing—review and editing, J.D.M.D.; supervision, J.D.M.D. All authors have read and agreed to the published version of the manuscript. [**TODO:** confirm contributions]

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Institutional Review Board Statement: Ethical review and approval were waived for this study owing to the retrospective, fully de-identified, secondary-use nature of the MIMIC-IV database. The original collection and de-identification of the data were reviewed and approved by the Institutional Review Board of the Beth Israel Deaconess Medical Center (BIDMC), which granted a waiver of informed consent for the creation and sharing of the database [9].

Informed Consent Statement: Patient consent was waived by the Institutional Review Board of the Beth Israel Deaconess Medical Center for the original collection of the MIMIC-IV database, owing to its retrospective and fully de-identified nature [9]. No participants were directly recruited by the authors.

Data Availability Statement: The clinical data used in this study are available from the MIMIC-IV database but restrictions apply to their availability: they were used under a data use agreement and are therefore not publicly redistributable. Data are available via PhysioNet [10] to any researcher who completes the required training and signs the PhysioNet Credentialed Health Data Use Agreement (https://physionet.org/content/mimiciv/). The SQL scripts and notebooks for cohort extraction, preprocessing and modelling are available at [**TODO:** insert public repository URL].

Acknowledgments: The authors thank the MIMIC-IV team at the Laboratory for Computational Physiology (MIT) and Beth Israel Deaconess Medical Center for maintaining and providing open access to the database. [**TODO:** add any acknowledgments, e.g., clinical reviewers of the diagnosis grouping]

Conflicts of Interest: The authors declare no conflicts of interest. [**TODO:** confirm; the first author is employed by a private company unrelated to this study—state this if the journal requires it]

Abbreviations

AP	Average Precision
AUC	Area Under the Receiver Operating Characteristic Curve
BIDMC	Beth Israel Deaconess Medical Center
CDC	Centers for Disease Control and Prevention
CV	Cross-Validation
ET	Extra Trees
FAD	Federally Available Data
HRSA	Health Resources and Services Administration
ICD	International Classification of Diseases
ICU	Intensive Care Unit
IQR	Interquartile Range
LR	Logistic Regression
MIMIC	Medical Information Mart for Intensive Care
ML	Machine Learning
PR	Precision–Recall
RF	Random Forest
ROC	Receiver Operating Characteristic
SMM	Severe Maternal Morbidity
SMOTE	Synthetic Minority Over-sampling Technique
XGB	XGBoost (Extreme Gradient Boosting)

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